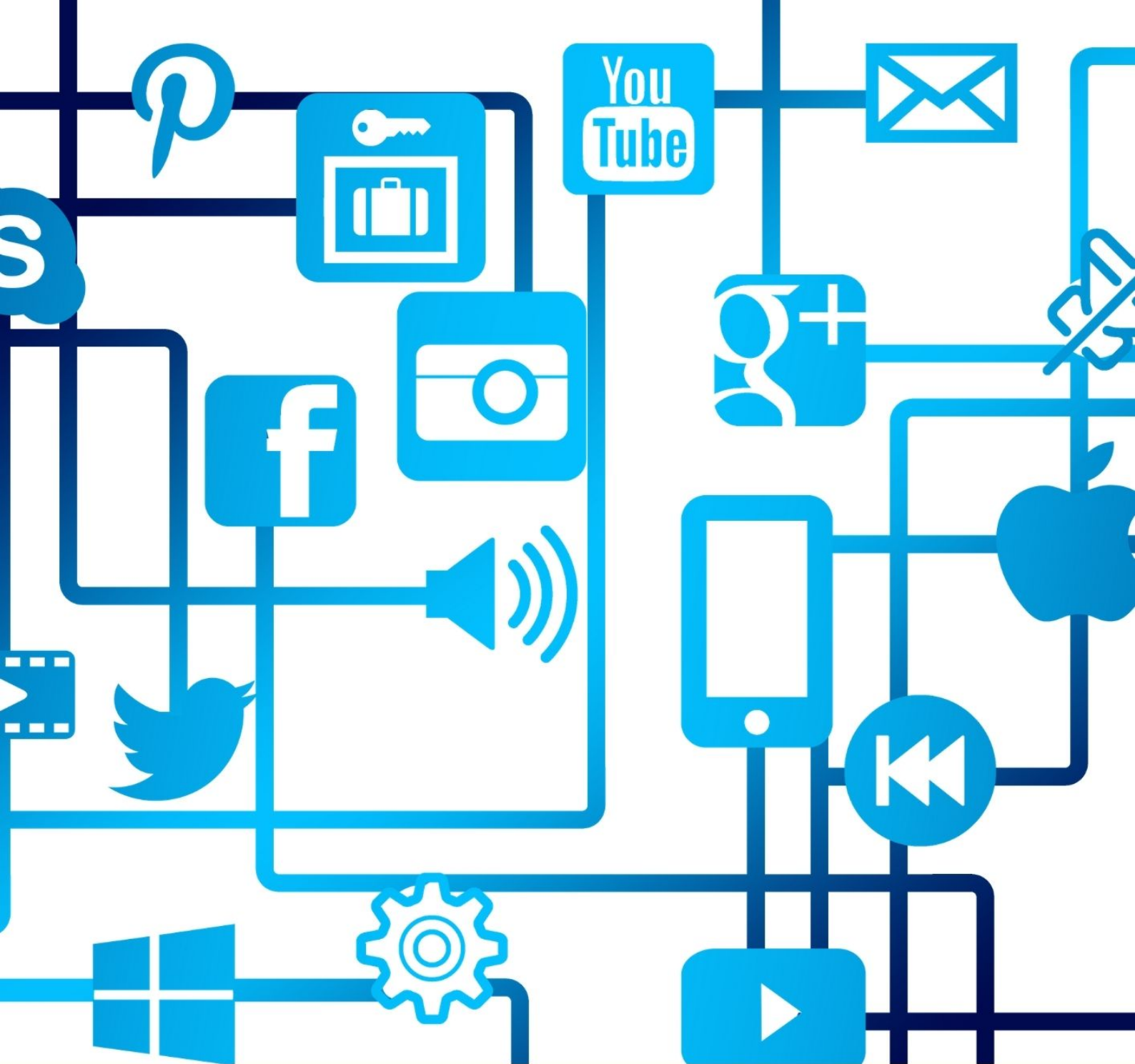




SOCIAL NETWORKS

Prof. Sudarshan Iyengar
Computer Science and Engineering
IIT Ropar



INDEX

S. No	Topic	Page No.
	<i>Week 1</i>	
1	Introduction	1
2	Answer to the puzzle	12
3	Introduction to Python-1	20
4	Introduction to Python-2	44
5	Introduction to Networkx-1	73
6	Introduction to Networkx-2	84
7	Social Networks: The Challenge	130
8	Google Page Rank	136
9	Searching in a Network	139
10	Link Prediction	142
11	The Contagions	145
12	Importance of Acquaintances	147
13	Marketing on Social Networks	149
	<i>Week 2</i>	
14	Introduction to Datasets	151
15	Ingredients Network	158
16	Synonymy Network	162
17	Web Graph	165
18	Social Network Datasets	168
19	Datasets: Different Formats	171
20	Datasets : How to Download?	185
21	Datasets: Analysing Using Networkx	199
22	Datasets: Analysing Using Gephi	227
23	Introduction : Emergence of Connectedness	240
24	Advanced Material : Emergence of Connectedness	245
25	Programming Illustration : Emergence of Connectedness	253
26	Summary to Datasets	274
	<i>Week 3</i>	
27	Introduction	276
28	Granovetter's Strength of weak ties	279
29	Triads, clustering coefficient and neighborhood overlap	283
30	Structure of weak ties, bridges, and local bridges	293

31	Validation of Granovetter's experiment using cell phone data	300
32	Embeddedness	304
33	Structural Holes	310
34	Social Capital	313
35	Finding Communities in a graph (Brute Force Method)	321
36	Community Detection Using Girvan Newman Algorithm	337
37	Visualising Communities using Gephi	345
38	Tie Strength, Social Media and Passive Engagement	359
39	Betweenness Measures and Graph Partitioning	367
40	Strong and Weak Relationship - Summary	376

Week 4

41	Introduction to Homophily - Should you watch your company ?	383
42	Selection and Social Influence	392
43	Interplay between Selection and Social Influence	402
44	Homophily - Definition and measurement	417
45	Foci Closure and Membership Closure	431
46	Introduction to Fatman Evolutionary model	435
47	Fatman Evolutionary Model- The Base Code (Adding people)	439
48	Fatman Evolutionary Model- The Base Code (Adding Social Foci)	454
49	Fatman Evolutionary Model- Implementing Homophily	470
50	Quantifying the Effect of Triadic Closure	480
51	Fatman Evolutionary Model- Implementing Closures	484
52	Fatman Evolutionary Model- Implementing Social Influence	500
53	Fatman Evolutionary Model- Storing and analyzing longitudinal data	512

Week 5

54	Spatial Segregation: An Introduction	524
55	Spatial Segregation: Simulation of the Schelling Model	531
56	Spatial Segregation: Conclusion	536
57	Schelling Model Implementation-1(Introduction)	538
58	Schelling Model Implementation-2 (Base Code)	543
59	Schelling Model Implementation-3 (Visualization and Getting a list of boundary and internal nodes)	557
60	Schelling Model Implementation-4 (Getting a list of unsatisfied nodes)	572
61	Schelling Model Implementation-5 (Shifting the unsatisfied nodes and visualizing the final graph)	580
62	Chapter - 5 Positive And Negative Relationships (Introduction)	587
63	Structural Balance	589

64	Enemy'S Enemy Is A Friend	597
65	Characterizing The Structure Of Balanced Networks	605
66	Balance Theorem	610
67	Proof Of Balance Theorem	616
68	Introduction to positive and negative edges	621
69	Outline of implemantation	627
70	Creating graph, displaying it and counting unstable triangles	634
71	Moving a network from an unstable to stable state	646
72	Forming two coalitions	667
73	Forming two coalitions contd	678
74	Visualizing coalitions and the evolution	682

Week 6

75	The Web Graph	688
76	Collecting the Web Graph	694
77	Equal Coin Distribution	703
78	Random Coin Dropping	709
79	Google Page Ranking Using Web Graph	715
80	Implementing PageRank Using Points Distribution Method-1	718
81	Implementing PageRank Using Points Distribution Method-2	723
82	Implementing PageRank Using Points Distribution Method-3	734
83	Implementing PageRank Using Points Distribution Method-4	747
84	Implementing PageRank Using Random Walk Method -1	754
85	Implementing PageRank Using Random Walk Method -2	757
86	DegreeRank versus PageRank	766

Week 7

87	We Follow	771
88	Why do we Follow?	777
89	Diffusion in Networks	784
90	Modeling Diffusion	787
91	Modeling Diffusion (continued)	793
92	Impact of Commmunities on Diffusion	798
93	Cascade and Clusters	809
94	Knowledge, Thresholds and the Collective Action	817
95	An Introduction to the Programming Screencast (Coding 4 major ideas)	824
96	The Base Code	830
97	Coding the First Big Idea - Increasing the Payoff	842

98	Coding the Second Big Idea - Key People	858
99	Coding the Third Big Idea- Impact of Communities on Cascades	864
100	Coding the Fourth Big Idea - Cascades and Clusters	879

Week 8

101	Introduction to Hubs and Authorities (A Story)	891
102	Principle of Repeated Improvement (A story)	895
103	Principle of Repeated Improvement (An example)	899
104	Hubs and Authorities	903
105	PageRank Revisited - An example	909
106	PageRank Revisited - Convergence in the Example	913
107	PageRank Revisited - Conservation and Convergence	917
108	PageRank, conservation and convergence - Another example	919
109	Matrix Multiplication (Pre-requisite 1)	923
110	Convergence in Repeated Matrix Multiplication (Pre-requisite 1)	930
111	Addition of Two Vectors (Pre-requisite 2)	937
112	Convergence in Repeated Matrix Multiplication- The Details	943
113	PageRank as a Matrix Operation	952
114	PageRank Explained	956

Week 9

115	Introduction to Powerlaw	962
116	Why do Normal Distributions Appear?	969
117	Power Law emerges in WWW graphs	974
118	Detecting the Presence of Powerlaw	981
119	Rich Get Richer Phenomenon	986
120	Summary So Far	993
121	Implementing Rich-getting-richer Phenomenon (Barabasi-Albert Model)-1	996
122	Implementing Rich-getting-richer Phenomenon (Barabasi-Albert Model)-2	1002
123	Implementing a Random Graph (Erdos- Renyi Model)-1	1028
124	Implementing a Random Graph (Erdos- Renyi Model)-2	1031
125	Forced Versus Random Removal of Nodes (Attack Survivability)	1041

Week 10

126	Rich Get Richer - A Possible Reason	1054
127	Rich Get Richer - The Long Tail	1063
128	Epidemics- An Introduction	1071
129	Introduction to epidemics (contd..)	1079
130	Simple Branching Process for Modeling Epidemics	1082

131	Simple Branching Process for Modeling Epidemics (contd..)	1090
132	Basic Reproductive Number	1095
133	Modeling epidemics on complex networks	1101
134	SIR and SIS spreading models	1106
135	Comparison between SIR and SIS spreading models	1120
136	Basic Reproductive Number Revisited for Complex Networks	1126
137	Percolation model	1131
138	Analysis of basic reproductive number in branching model (The problem statement)	1137
139	Analyzing basic reproductive number 2	1141
140	Analyzing basic reproductive number 3	1148
141	Analyzing basic reproductive number 4	1152
142	Analyzing basic reproductive number 5	1156

Week 11

143	Small World Effect - An Introduction	1160
144	Milgram's Experiment	1168
145	The Reason	1174
146	The Generative Model	1180
147	Decentralized Search - I	1183
148	Decentralized Search - II	1188
149	Decentralized Search - III	1196

Week 12

150	Programming illustration- Small world networks : Introduction	1209
151	Base code	1218
152	Making homophily based edges	1221
153	Adding weak ties	1234
154	Plotting change in diameter	1238
155	Programming illustration- Myopic Search : Introduction	1241
156	Myopic Search	1247
157	Myopic Search comparison to optimal search	1263
158	Time Taken by Myopic Search	1269
159	PseudoCores : Introduction	1281
160	How to be Viral	1286
161	Who are the right key nodes?	1290
162	finding the right key nodes (the core)	1292
163	Coding K-Shell Decomposition	1303
164	Coding cascading Model	1315

165	Coding the importance of core nodes in cascading	1330
166	Pseudo core	1343